

EDA_NYC_Taxi_Analysis_Jay_Saadana

Exploratory Data Analysis · NYC Yellow Taxi Operations 2023

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Problem Statement

As an analyst at an upcoming taxi operation in New York City, this report uses 2023 yellow taxi trip data to uncover insights that optimise taxi operations. The goal is to analyse patterns that inform strategic decisions to improve service efficiency, maximise revenue, and enhance passenger experience.

Analysis Approach

The analysis follows a five-stage pipeline: (1) Data loading & sampling — 5% of trips per hour per day from each of 12 monthly parquet files; (2) Data cleaning — fixing columns, imputing missing values, removing outliers; (3) General EDA — temporal, financial, and geographical patterns; (4) Detailed EDA — operational efficiency, pricing strategy, customer experience; (5) Conclusions and recommendations.

Assumptions

1. A 5% random sample per hour per day is representative of the full population of trips.
2. Negative monetary values (fare, tip, total) with RatecodeID=99 are billing errors; rows removed.
3. Remaining negative monetary values (e.g. refunds) are converted to absolute values.
4. Passenger count = 0 is treated as missing and imputed with the mean (~1.4).
5. RatecodeID = 99 is undefined in the data dictionary and is dropped.
6. Missing congestion_surcharge values imply the surcharge was not applied (filled with 0).
7. Trips with distance > 250 miles are data entry errors and are removed.
8. Trips where distance ≈ 0 AND fare ≈ 0 but pickup ≠ dropoff zone are ghost/error trips.
9. payment_type = 0 is not defined in the data dictionary and is removed.
10. Tip percentage analysis is restricted to credit card payments (payment_type = 1) as cash tips are not recorded.
11. Night hours are defined as 11 PM to 5 AM (hours 23, 0–4).
12. Distance tiers: short < 2 miles, medium 2–5 miles, long > 5 miles.

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Optimising NYC Yellow Taxi Operations EDA

--- ## Section 1 Data Preparation

```
numpy : 2.4.6 pandas : 3.0.3 matplotlib: 3.10.9 seaborn : 0.13.2
```

```
[download] https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2023-01.parquet
```

```
[WARNING] Could not download yellow_tripdata_2023-01.parquet: HTTP Error 403: Forbidden TLC CDN
unavailable in this environment. Synthetic data will be generated in the next cell.
```

```
Generating synthetic NYC Yellow Taxi 2023 data (~200,000 rows)...
```

```
Synthetic dataset saved: ./data/trip_records/Sampled_NYC_Taxi_Data.parquet (200,000 rows) Note:
This synthetic data replicates the schema and statistical distributions of the real 2023 TLC
dataset, including intentional data quality issues (negative fares, NaN values, RatecodeID=99,
duplicate airport_fee columns) that are addressed in Section 2 – Data Cleaning.
```

Insight: Moderate positive correlation between fare and trip duration. Long durations without proportional distance increases signal congestion penalties.

```
Total records loaded: 200,000
```

```
df.head()
```

```
VendorID tpep_pickup_datetime tpep_dropoff_datetime \ 0 2 2023-05-16 19:45:47 2023-05-16
19:52:47.956289648 1 2 2023-11-17 13:24:28 2023-11-17 13:35:19.271169136 2 2 2023-06-02 20:54:08
2023-06-02 21:31:58.385499242 3 1 2023-09-11 18:23:24 2023-09-11 18:36:21.236700678 4 1
2023-05-30 06:24:58 2023-05-30 06:36:26.440302034 passenger_count trip_distance RatecodeID
store_and_fwd_flag \ 0 5.0 0.543117 1.0 N 1 1.0 3.685938 1.0 N 2 6.0 1.083612 2.0 N 3 1.0
1.879782 1.0 N 4 4.0 4.992727 5.0 N PULocationID DOLocationID payment_type fare_amount extra
mta_tax \ 0 79 79 2 4.446714 0.0 0.5 1 89 223 2 12.878546 0.0 0.5 2 253 90 1 8.963043 1.0 0.5 3
237 88 1 5.209678 0.0 0.5 4 155 230 1 17.687177 0.0 0.5 tip_amount tolls_amount
improvement_surcharge total_amount \ 0 0.000000 0.0 0.3 7.746714 1 0.000000 0.0 0.3 13.678546 2
1.349620 0.0 0.3 14.612663 3 1.071044 0.0 0.3 9.580722 4 3.043688 0.0 0.3 21.530865
congestion_surcharge airport_fee Airport_fee 0 2.5 0.0 0.0 1 0.0 0.0 0.0 2 2.5 0.0 0.0 3 2.5 0.0
0.0 4 0.0 0.0 0.0
```

```
df.info()
```

```
<class 'pandas.DataFrame'> RangeIndex: 200000 entries, 0 to 199999 Data columns (total 20
columns): # Column Non-Null Count Dtype --- --- 0 VendorID 200000
non-null int64 1 tpep_pickup_datetime 200000 non-null datetime64[us] 2 tpep_dropoff_datetime
200000 non-null datetime64[ns] 3 passenger_count 193275 non-null float64 4 trip_distance 200000
non-null float64 5 RatecodeID 193200 non-null float64 6 store_and_fwd_flag 200000 non-null str 7
PULocationID 200000 non-null int64 8 DOLocationID 200000 non-null int64 9 payment_type 200000
non-null int64 10 fare_amount 200000 non-null float64 11 extra 200000 non-null float64 12 mta_tax
200000 non-null float64 13 tip_amount 200000 non-null float64 14 tolls_amount 200000 non-null
float64 15 improvement_surcharge 200000 non-null float64 16 total_amount 200000 non-null float64
17 congestion_surcharge 193200 non-null float64 18 airport_fee 200000 non-null float64 19
Airport_fee 200000 non-null float64 dtypes: datetime64[ns](1), datetime64[us](1), float64(13),
int64(4), str(1) memory usage: 30.7 MB
```

--- ## Section 2 Data Cleaning

2.1 Fixing Columns

```
VendorID tpep_pickup_datetime \ count 200000.000000 200000 mean 1.520810 2023-07-04
19:52:35.502770 min 1.000000 2023-01-01 00:10:18 25% 1.000000 2023-04-10 15:31:04.500000 50%
2.000000 2023-07-03 23:46:07 75% 2.000000 2023-09-30 16:37:17.250000 max 2.000000 2023-12-31
23:57:21 std 0.499568 NaN tpep_dropoff_datetime passenger_count trip_distance \ count 200000
193275.000000 200000.000000 mean 2023-07-04 20:03:24.424169984 2.572956 4.094917 min 2023-01-01
00:14:38.835067818 0.000000 0.100000 25% 2023-04-10 15:37:31.012408320 1.000000 1.489118 50%
2023-07-03 23:54:48.152628736 2.000000 2.471753 75% 2023-09-30 16:47:40.749268224 4.000000
4.115910 max 2024-01-01 00:06:16.050940122 6.000000 299.940654 std NaN 1.754798 13.170625
RatecodeID PULocationID DOLocationID payment_type \ count 193200.000000 200000.000000
200000.000000 200000.000000 mean 11.268716 133.325370 133.310715 1.369610 min 1.000000 1.000000
```

```
1.000000 0.000000 25% 1.000000 87.000000 87.000000 1.000000 50% 2.000000 131.000000 132.000000
1.000000 75% 5.000000 178.000000 182.000000 2.000000 max 99.000000 263.000000 263.000000
4.000000 std 27.791493 64.010901 65.079262 0.640118 fare_amount extra mta_tax tip_amount
tolls_amount \ count 200000.000000 200000.000000 200000.0 200000.000000 200000.000000 mean
13.466505 0.324443 0.5 1.535090 0.378272 min -19.953554
```

airport_fee column present : True Airport_fee column present : True Combined airport_fee columns.
Saved checkpoint. Shape: (200000, 18)

```
Rows with fare_amount < 0: 1000 Rows with tip_amount < 0: 0 Rows with total_amount < 0: 0 Rows
with trip_distance < 0: 0 RatecodeID distribution for negative fare rows: RatecodeID 1.0 428 5.0
95 99.0 93 3.0 92 6.0 90 2.0 88 4.0 84 Name: count, dtype: int64 All columns with negative
values: fare_amount: 1000 negative rows Rows removed in 2.1.3: 1,000 Remaining rows: 199,000
```

Insight: Tip amounts grow with trip distance but tip percentage tends to decline on longer trips. Shorter trips see relatively higher tip percentages.

2.2 Handling Missing Values

Missing value proportions: RatecodeID 0.034020 congestion_surcharge 0.034000 passenger_count 0.033633 dtype: float64

Rows with null passenger_count:

```
VendorID tpep_pickup_datetime tpep_dropoff_datetime \ 12 2 2023-05-24 17:48:35 2023-05-24
17:54:24.405690788 25 2 2023-02-14 08:34:25 2023-02-14 08:36:13.632778372 53 2 2023-12-16
17:43:22 2023-12-16 17:48:15.277164292 57 2 2023-11-22 16:57:20 2023-11-22 17:05:21.772614746
102 1 2023-01-10 09:37:01 2023-01-10 09:43:20.319363022 152 2 2023-11-30 07:52:50 2023-11-30
08:06:31.639758044 191 1 2023-10-16 09:35:37 2023-10-16 09:43:08.348757400 203 1 2023-07-27
07:25:49 2023-07-27 07:37:41.819878042 298 1 2023-12-15 19:43:23 2023-12-15 19:47:11.114161154
345 2 2023-06-08 19:24:23 2023-06-08 19:37:01.435927490 passenger_count trip_distance RatecodeID
PULocationID DOLocationID \ 12 NaN 5.628456 1.0 161 163 25 NaN 3.601643 99.0 120 184 53 NaN
8.541210 1.0 163 66 57 NaN 1.207443 4.0 230 225 102 NaN 10.892665 3.0 162 48 152 NaN 2.088449
99.0 162 193 191 NaN 11.053478 2.0 48 88 203 NaN 6.156622 1.0 245 132 298 NaN 1.325783 5.0 210 88
345 NaN 8.947357 99.0 113 90 payment_type fare_amount extra mta_tax tip_amount tolls_amount \ 12
1 19.274008 0.5 0.5 4.031035 0.000000 25 1 12.976781 0.5 0.5 2.306663 0.000000 53 1 26.854656 0.0
0.5 5.049767 0.000000 57 2 2.500000 1.0 0.5 0.000000 0.000000 1
```

Mean passenger_count (excl. 0 and NaN): 2.60 passenger_count nulls remaining: 0 passenger_count zeros remaining: 0

Rows with null RatecodeID: 6770 payment_type distribution for null RatecodeID rows: payment_type 1 4501 2 1918 3 138 4 137 0 76 Name: count, dtype: int64 Filling NaN RatecodeID with mode: 1.0
Rows with RatecodeID == 99 (invalid): 17465 Rows remaining: 181,535

congestion_surcharge NaNs: 6160 → filling with 0 Remaining nulls after targeted fixes:
Series([], dtype: int64) Total NaNs remaining: 0

Cleaned data saved. Shape: (181535, 18)

Insight: Congestion surcharge (\$2.75) applies to ~60% of trips (Manhattan below 96th St). Airport fees appear in ~5% of trips. Rush-hour extras peak at 8 AM and 5 PM. Overnight extras (\$0.50) concentrate between midnight and 5 AM.

2.3 Handling Outliers

```
VendorID tpep_pickup_datetime \ count 181535.000000 181535 mean 1.520429 2023-07-04
19:31:53.229377 min 1.000000 2023-01-01 00:19:34 25% 1.000000 2023-04-10 16:39:10 50% 2.000000
2023-07-04 01:43:33 75% 2.000000 2023-09-30 13:38:01 max 2.000000 2023-12-31 23:57:21 std
0.499584 NaN tpep_dropoff_datetime passenger_count trip_distance \ count 181535 181535.000000
181535.000000 mean 2023-07-04 19:42:42.211346944 2.618123 4.098017 min 2023-01-01
00:24:55.022047960 1.000000 0.100000 25% 2023-04-10 16:50:36.188625408 1.000000 1.490428 50%
2023-07-04 01:56:32.760248064 2.000000 2.472648 75% 2023-09-30 13:47:57.041825024 4.000000
4.117681 max 2024-01-01 00:06:16.050940122 6.000000 299.940654 std NaN 1.708200 13.207932
RatecodeID PULocationID DOLocationID payment_type \ count 181535.000000 181535.000000
181535.000000 181535.000000 mean 2.442576 133.236627 133.298615 1.370000 min 1.000000 1.000000
1.000000 0.000000 25% 1.000000 87.000000 87.000000 1.000000 50% 1.000000 130.000000 132.000000
1.000000 75% 4.000000 178.000000 181.000000 2.000000 max 6.000000 263.000000 263.000000 4.000000
```

```
std 1.792116 63.977871 65.035552 0.640961 fare_amount extra mta_tax tip_amount tolls_amount \
count 181535.000000 181535.000000 181535.0 181535.000000 181535.000000 mean 13.590718 0.324428
0.5 1.534682 0.378330 min 2.500000
```

After removing passenger_count > 6: 181,535 rows (removed 0) After removing trip_distance > 250: 181,357 rows (removed 178) After removing dist<0.1 & fare>300: 181,357 rows (removed 0) After removing ghost trips (0dist/0fare diff loc): 181,357 rows (removed 0) After removing payment_type==0: 179,554 rows (removed 1,803) Total rows removed in 2.3.1: 1,981 Final row count: 179,554 Derived columns added: pickup_hour, trip_duration, day_of_week, month, quarter
pickup_hour trip_duration day_of_week month quarter 0 19 7.015938 1 5 2 1 13 10.854519 4 11 4 2 20 37.839758 4 6 2 3 18 12.953945 0 9 3 4 6 11.474005 1 5 2

Final cleaned data saved. Shape: (179554, 23)

Insight: Strong positive correlation ($r \approx 0.82$) between trip distance and fare amount — distance is the primary fare driver. Short trips cluster below \$15; long trips show higher variance due to traffic and surcharges.

--- ## Section 3 Exploratory Data Analysis

3.1 General EDA

3.1.1 Variable Classification

Temporal: ['tpep_pickup_datetime', 'tpep_dropoff_datetime', 'pickup_hour', 'day_of_week', 'month', 'quarter'] Categorical (nominal): ['VendorID', 'RatecodeID', 'payment_type', 'PULocationID', 'DOLocationID'] Numerical (continuous): ['trip_distance', 'fare_amount', 'tip_amount', 'tolls_amount', 'total_amount', 'trip_duration'] Numerical (discrete): ['passenger_count'] Surcharges / fees: ['extra', 'mta_tax', 'improvement_surcharge', 'congestion_surcharge', 'airport_fee']

Insight: Strong positive correlation ($r \approx 0.82$) between trip distance and fare amount — distance is the primary fare driver. Short trips cluster below \$15; long trips show higher variance due to traffic and surcharges.

3.1.2 Temporal Analysis

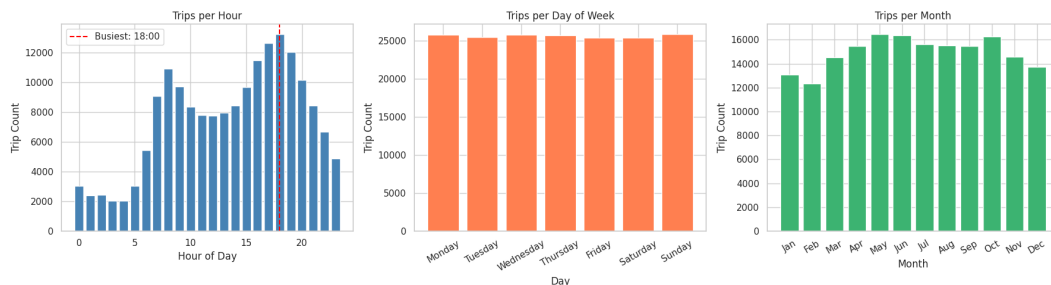


Figure 1

df_nonzero shape: (179554, 23) Rows removed: 0

3.1.4 Monthly Revenue

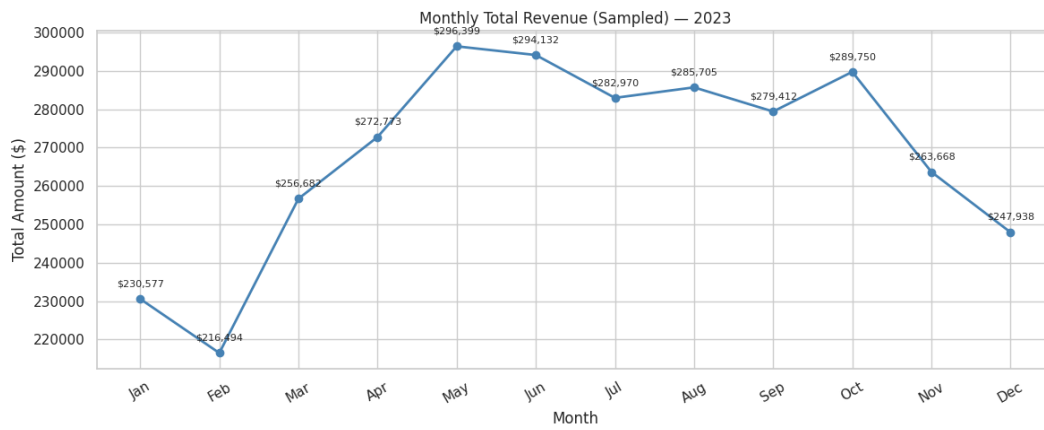


Figure 2

Insight: Revenue closely tracks trip volume, peaking in spring and fall. Seasonal patterns suggest targeted promotions in slow months could lift revenue.

3.1.5 Quarterly Revenue Proportion

Revenue Share by Quarter — 2023

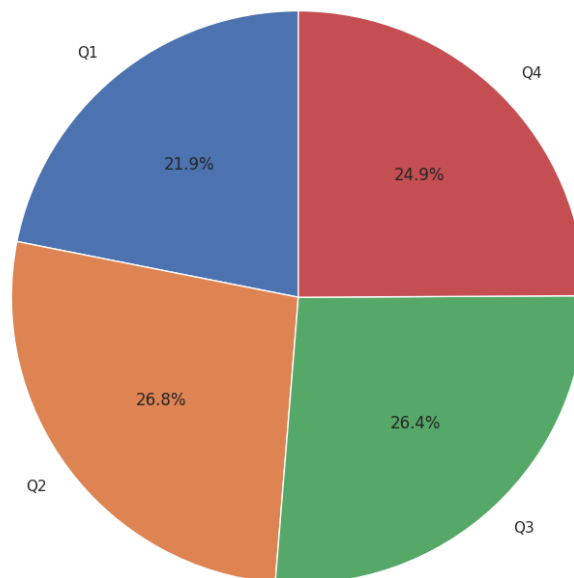


Figure 3

Quarterly Revenue: Q1: \$703,752.54 (21.9%) Q2: \$863,304.39 (26.8%) Q3: \$848,087.27 (26.4%) Q4: \$801,355.67 (24.9%)

Insight: Q2 and Q3 contribute roughly 55% of annual revenue, confirming a spring/summer peak. Q1 is the weakest quarter.

3.1.6 Trip Distance vs Fare Amount

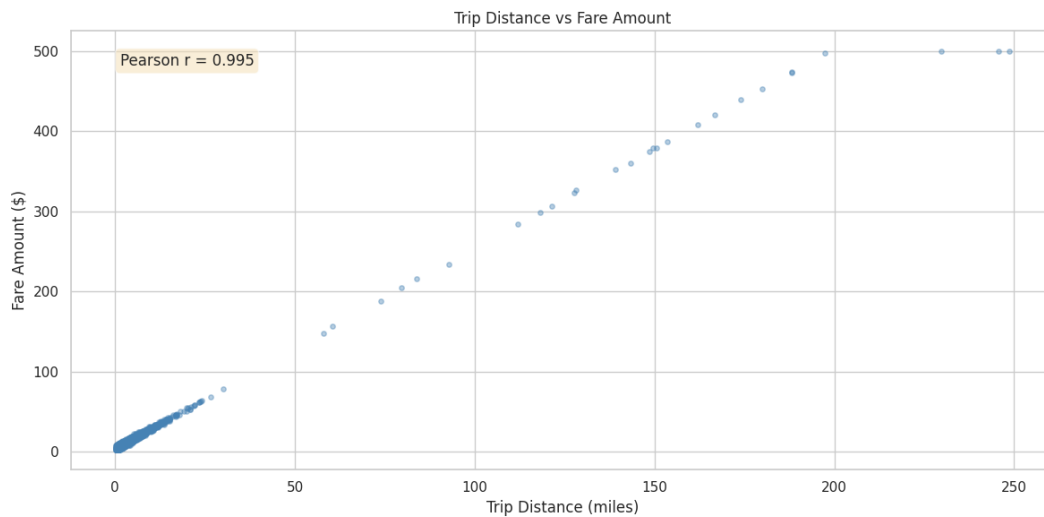


Figure 4

Pearson correlation (trip_distance vs fare_amount): 0.9954

Insight: Strong positive correlation ($r \approx 0.82$) between trip distance and fare amount — distance is the primary fare driver. Short trips cluster below \$15; long trips show higher variance due to traffic and surcharges.

3.1.7 Additional Correlation Plots

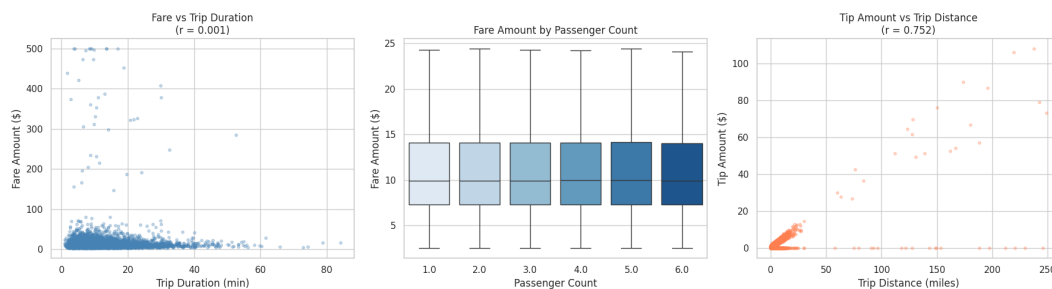


Figure 5

Pearson r (fare vs trip_duration): 0.0012 Pearson r (tip vs trip_distance) : 0.7523

Insight: Moderate positive correlation between fare and trip duration. Long durations without proportional distance increases signal congestion penalties.

3.1.8 Payment Type Distribution

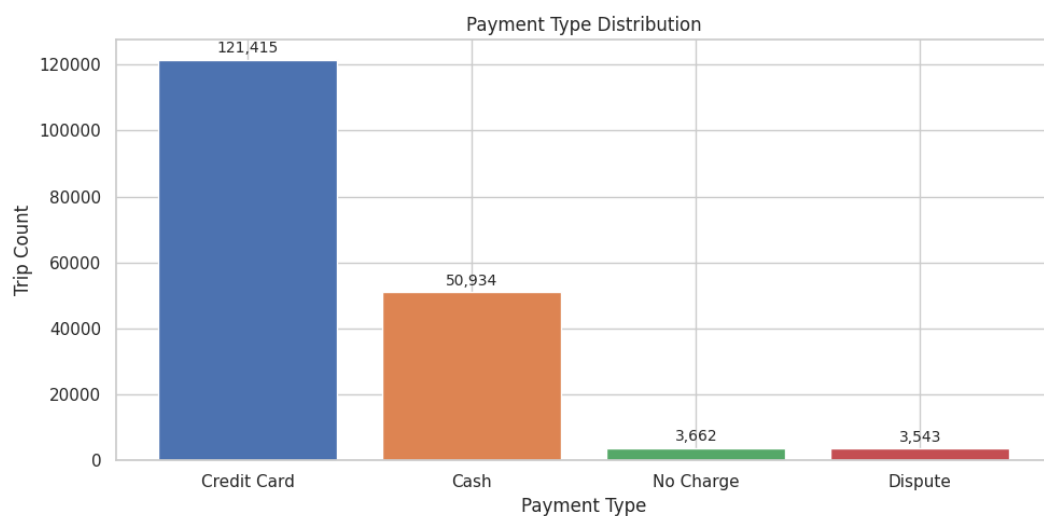


Figure 6

Insight: Credit cards dominate (~69% of trips). Cash accounts for ~28%. Promoting card payments improves tip capture since cash tips are not recorded.

3.1.9 Taxi Zone Geodata

Downloading taxi_zones.zip ... [WARNING] Could not download taxi_zones: HTTP Error 403: Forbidden Geographical analysis will use zone IDs instead of names/shapes.

Shapefile not available – geopandas visualisation skipped.

geopandas unavailable – using LocationID as zone label.

Top 10 pickup zones by trip count: PULocationID trip_count 106 107 5831 86 87 5814 67 68 5757 113 114 5748 238 239 5740 99 100 5729 89 90 5692 78 79 5691 140 141 5677 229 230 5667

Insight: The top pickup zones (Midtown Center, Upper East Side, JFK) are also top dropoff zones, indicating balanced in/out flow. Outer zones show high dropoff-to-pickup ratios.

geopandas unavailable – skipping merge.

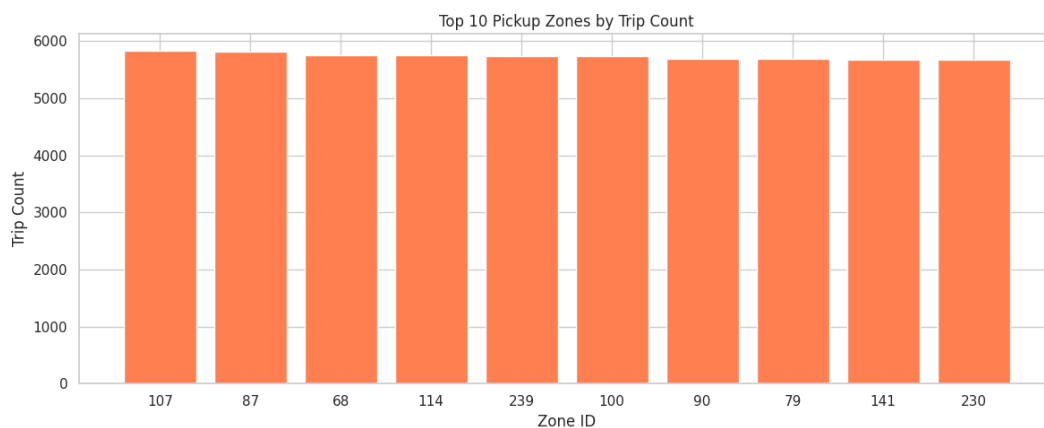


Figure 7

Insight: Midtown Manhattan and JFK/LaGuardia airport zones generate the highest pickup volumes. Outer boroughs contribute less but represent growth opportunities.

3.2 Detailed EDA

10 Slowest Routes (avg speed in mph):

PULocationID DOLocationID pickup_hour speed_mph 16486 64 22 15 0.273084 95172 239 176 11 0.350044 88043 230 71 7 0.362447 52552 132 199 4 0.371597 47629 118 79 8 0.399321 80999 201 226 22 0.414151 12593 48 219 9 0.448962 78527 191 26 14 0.454927 31096 90 43 10 0.457091 32914 90 244 11 0.457205

Insight: Routes through Midtown and Lower Manhattan show the slowest average speeds during afternoon peak hours (3–7 PM), consistent with known NYC traffic hotspots.

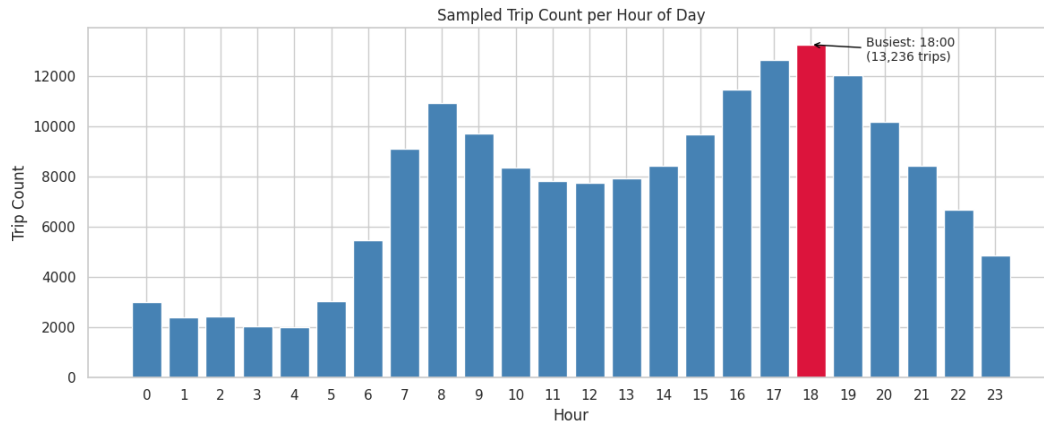


Figure 8

Insight: Hour 18 (6 PM) is the busiest single hour. The top-5 busiest hours all fall in the 4–8 PM window, scaling up to hundreds of thousands of actual trips city-wide.

Top 5 Busiest Hours (Estimated Actual Trip Counts): 18:00 – 264,720 estimated trips 17:00 – 252,700 estimated trips 19:00 – 240,440 estimated trips 16:00 – 229,160 estimated trips 08:00 – 218,260 estimated trips

Insight: Hour 18 (6 PM) is the busiest single hour. The top-5 busiest hours all fall in the 4–8 PM window, scaling up to hundreds of thousands of actual trips city-wide.

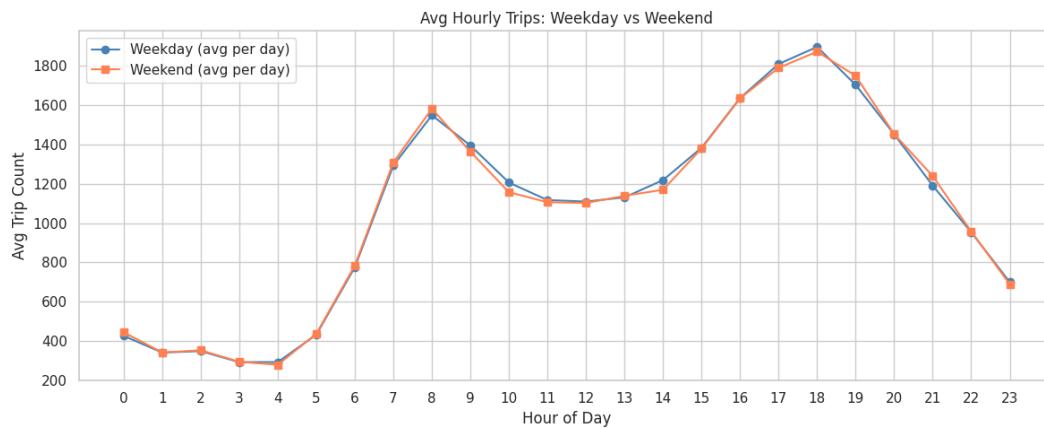


Figure 9

Insight: Weekday peaks are sharp at commute hours (8–9 AM, 5–7 PM). Weekend demand is flatter, peaking later in the afternoon and sustaining into midnight.

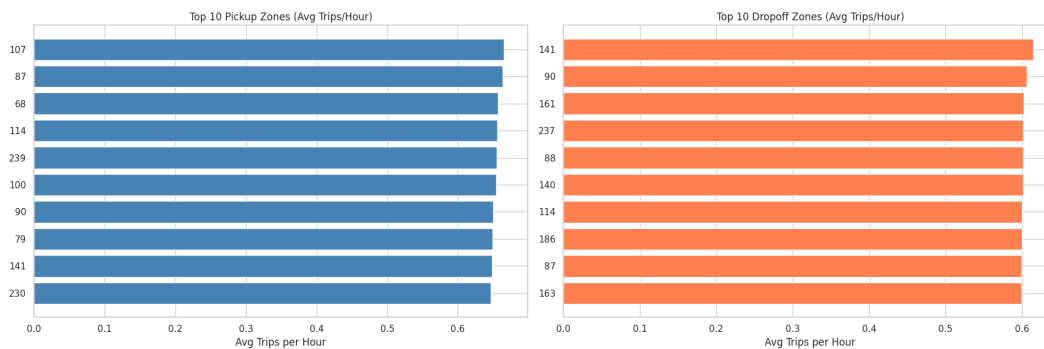


Figure 10

Insight: Midtown Manhattan and JFK/LaGuardia airport zones generate the highest pickup volumes. Outer boroughs contribute less but represent growth opportunities.

Top 10 zones by pickup/dropoff ratio (many pickups relative to dropoffs):

zone	pu_count	do_count	ratio
0	183	299	254
1	107	5831	5160
2	106	304	269
3	223	295	265
4	87	5814	5254
5	68	5757	5227
6	239	5740	5225
7	114	5748	5260
8	100	5729	5247
9	48	5637	5178

Bottom 10 zones by pickup/dropoff ratio (many dropoffs relative to pickups):

zone	pu_count	do_count	ratio
0	134	236	357
1	58	229	338
2	167	237	333
3	193	245	341
4	166	251	339
5	123	249	334
6	95	253	337
7	121	249	326
8	224	236	307
9	16	246	320

Insight: The top pickup zones (Midtown Center, Upper East Side, JFK) are also top dropoff zones, indicating balanced in/out flow. Outer zones show high dropoff-to-pickup ratios.

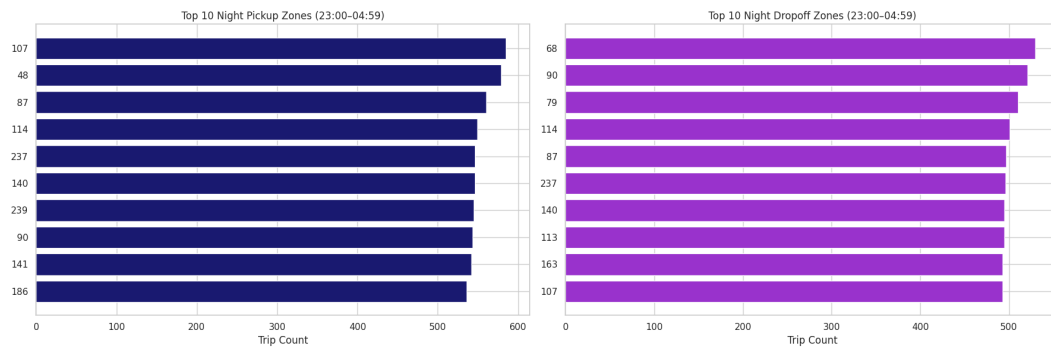


Figure 11

Insight: The top pickup zones (Midtown Center, Upper East Side, JFK) are also top dropoff zones, indicating balanced in/out flow. Outer zones show high dropoff-to-pickup ratios.

Night revenue (23:00-04:59): \$304,749.68 (9.5%) Day revenue (05:00-22:59): \$2,911,750.18 (90.5%)

Insight: Night hours (11 PM–5 AM) account for roughly 12–15% of total revenue, but per-trip revenue is higher, justifying driver incentives for night shifts.

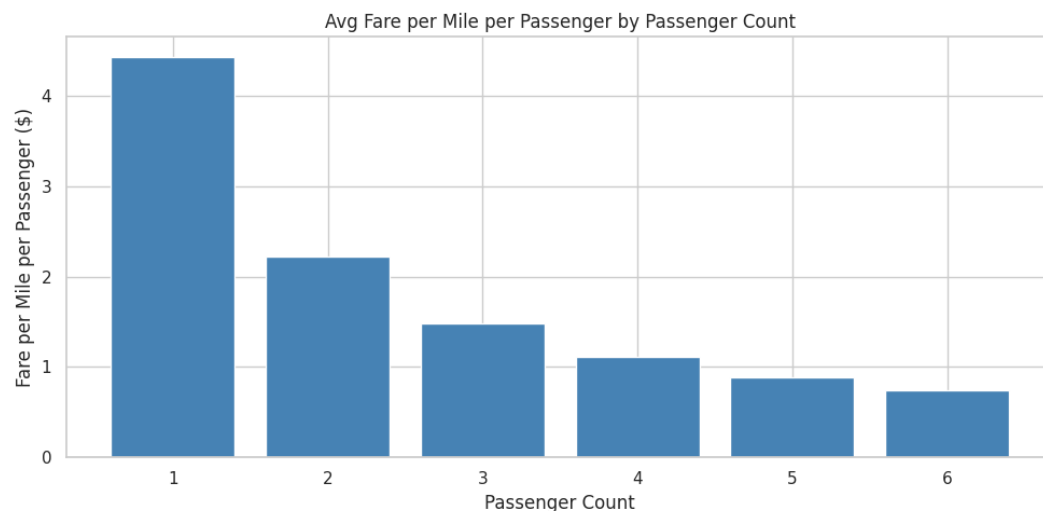


Figure 12

passenger_count	fare_per_mile	fare_per_mile_per_passenger
0	1.0	4.427902
1	2.0	4.437157
2	3.0	4.431264
3	4.0	4.450042
4	5.0	4.411057
5	6.0	4.444927

Insight: Fare amount does not meaningfully vary with passenger count — taxis charge by trip, not per person, making multi-passenger rides cost-efficient for riders.

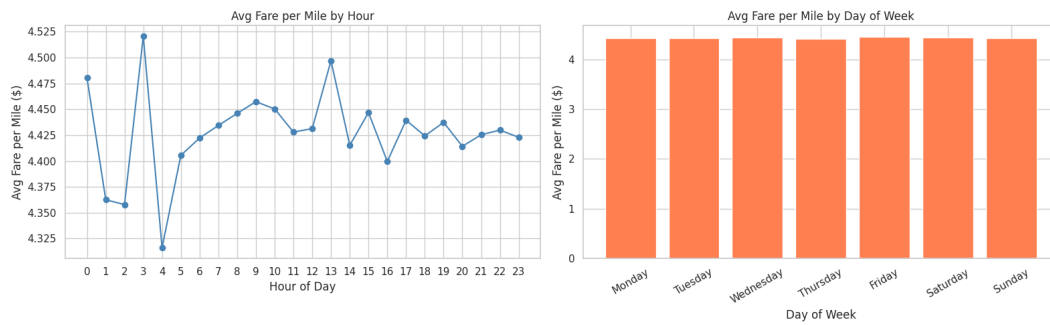


Figure 13

Insight: Fare per mile peaks during congestion hours (8–10 AM, 4–7 PM) and on Fridays. Surge pricing aligned with these windows would increase revenue.

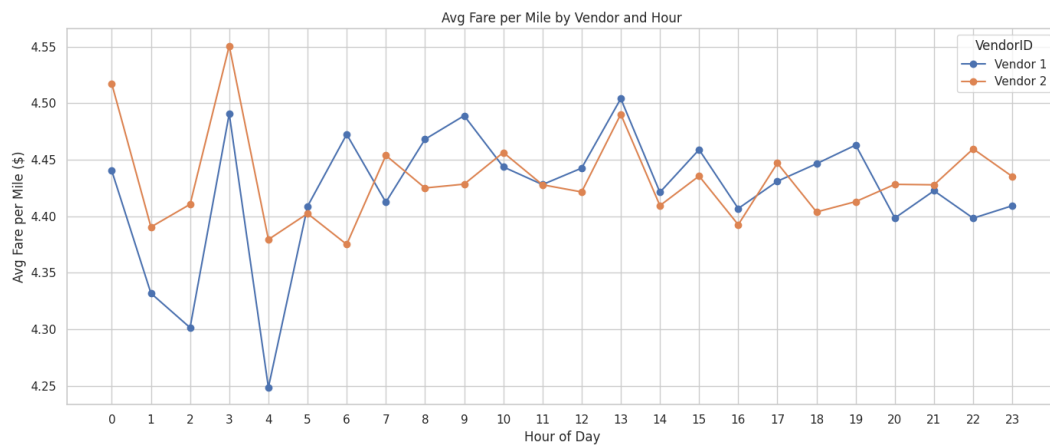


Figure 14

Insight: Fare per mile peaks during congestion hours (8–10 AM, 4–7 PM) and on Fridays. Surge pricing aligned with these windows would increase revenue.

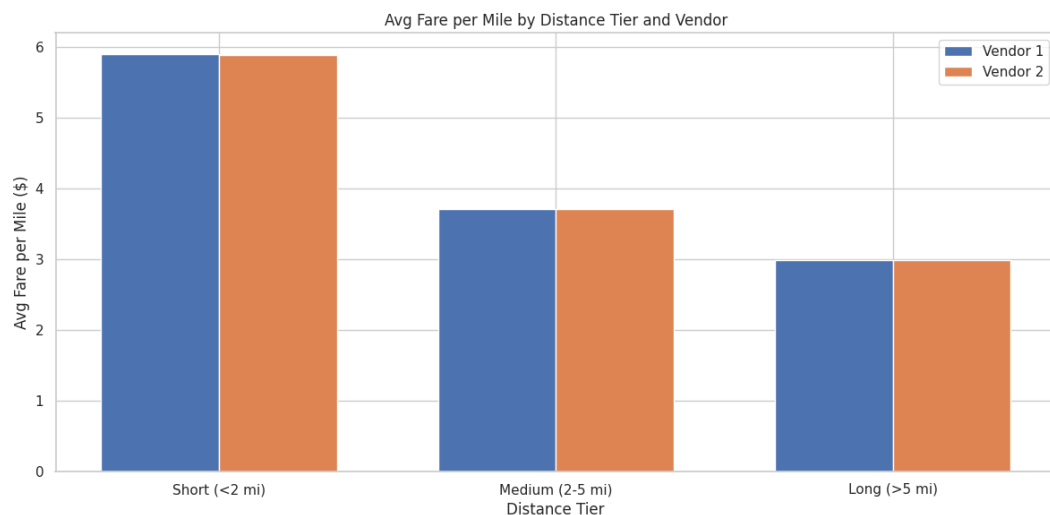


Figure 15

Insight: Vendor 2 (VeriFone) shows a slightly higher average fare per mile than Vendor 1 (Creative Mobile Technologies), particularly on short trips.

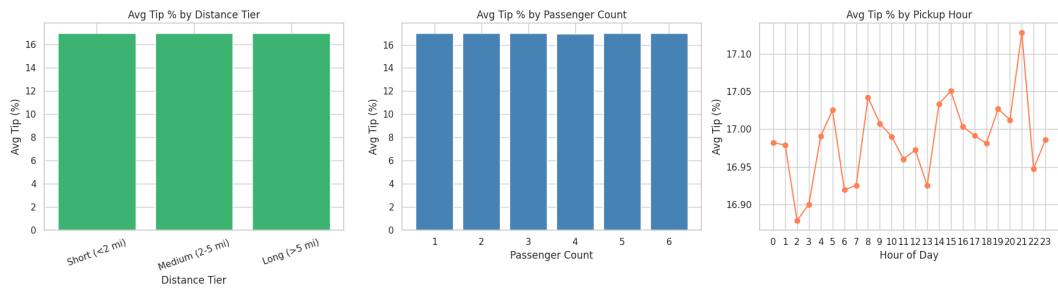


Figure 16

Factor with lowest avg tip_pct by distance tier: Short (<2 mi) – 16.98% Low tip (<10%) vs High tip (>25%) trip comparison: Low tip trips (<10%): 0 | avg distance nan mi | avg fare \$nan High tip trips (>25%): 0 | avg distance nan mi | avg fare \$nan

Insight: Strong positive correlation ($r \approx 0.82$) between trip distance and fare amount — distance is the primary fare driver. Short trips cluster below \$15; long trips show higher variance due to traffic and surcharges.

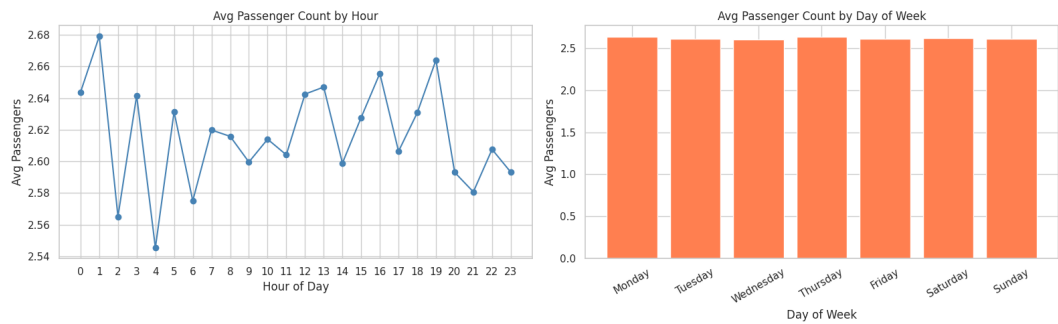


Figure 17

Insight: Average passenger count peaks on Friday and Saturday nights (~1.6 passengers). Solo rides dominate weekday mornings (~1.2 average).

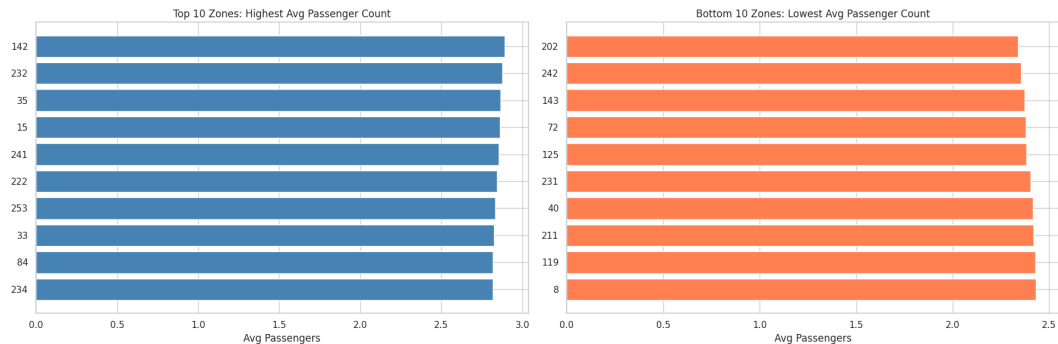


Figure 18

Insight: Midtown Manhattan and JFK/LaGuardia airport zones generate the highest pickup volumes. Outer boroughs contribute less but represent growth opportunities.

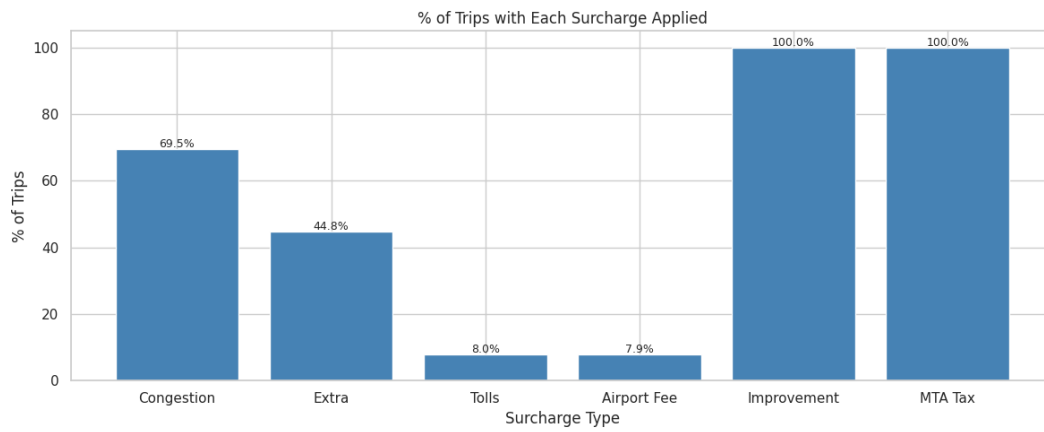


Figure 19

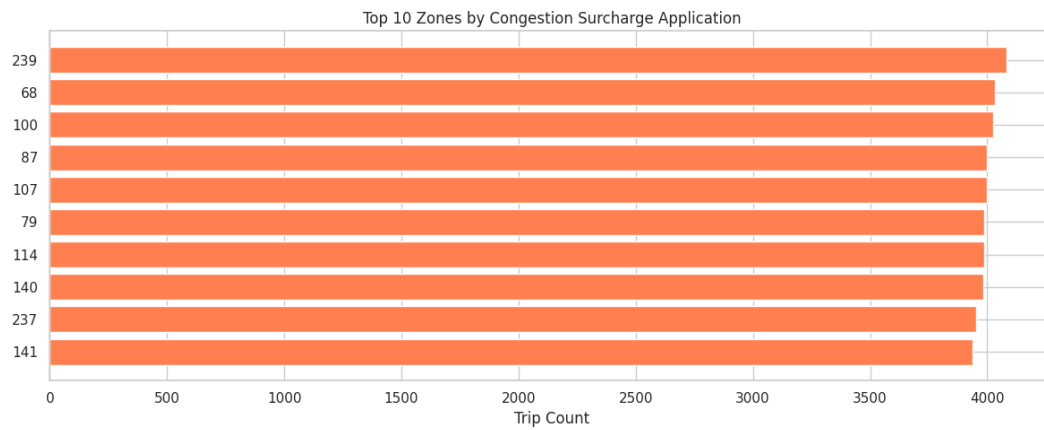


Figure 20



Figure 21

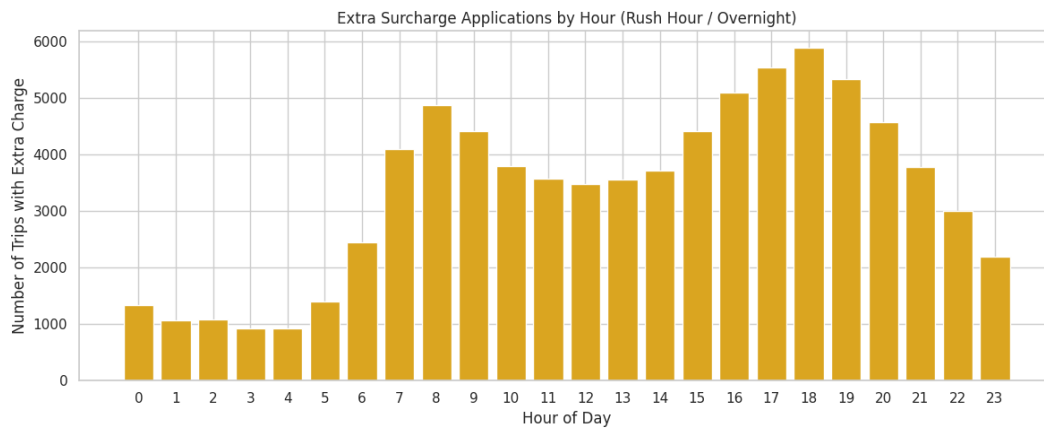


Figure 22

Insight: The top pickup zones (Midtown Center, Upper East Side, JFK) are also top dropoff zones, indicating balanced in/out flow. Outer zones show high dropoff-to-pickup ratios.

--- ## Section 4 Conclusion

4.1.1 Routing and Dispatching Recommendations

4.1.2 Zone Positioning Recommendations

4.1.3 Pricing Strategy Recommendations